

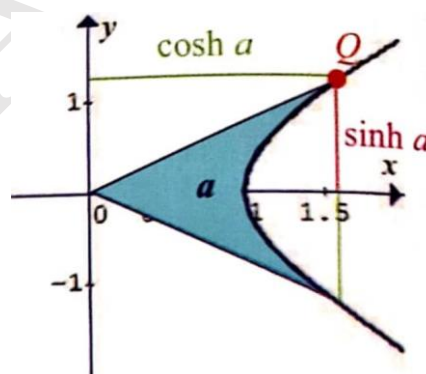
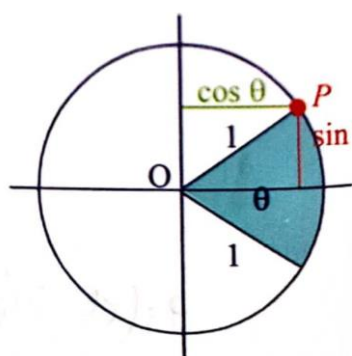
FP2 Chapter 2 Hyperbolic Functions

Lecture 2 Hyperbolic Functions

(Exponential form, Logarithmic form, Graphs, Identities, Osborn's Rule, Problem Solving)

Introduction:

- The trigonometric function $\sin x, \cos x, \tan x$ are called circular functions because they are defined in terms of a circle. A general point P , on the circumference of a circle $x^2 + y^2 = 1$ has coordinates $(\cos \theta, \sin \theta)$ where θ is the angle between the radius OP and the x -axis.
- In the 1760's, Vincenzo Riccati and Johann Heinrich Lambert working independently, tried the same thing with the hyperbola $x^2 - y^2 = 1$. They defined the coordinate of the point Q on the hyperbola (where $x > 0$) as $(\cosh a, \sinh a)$, adding 'h' to indicate hyperbola.
- In this case, the parameter a , is an area. The area is the shaded area form with the ray OQ , its reflection in the x -axis and the hyperbola.



- Notice, however, that if you formed the equivalent area in the case of a circle, that would have an area of θ . Thus they are really looking at the same thing.
- This is where hyperbolic function come from. In fact, we shall see, they can be expressed in terms of exponential functions.

Functions	Pronunciation
$\sinh$	shine
$\cosh$	cosh (coshine)
$\tanh$	than / tank
sech	shec
cosech	coshec
$\coth$	coth



Definitions and Graphs of Hyperbolic Functions

- The hyperbolic cosine of x is defined as

$$\cosh x = \frac{1}{2}(e^x + e^{-x}).$$

- The hyperbolic sine of x is defined as

$$\sinh x = \frac{1}{2}(e^x - e^{-x}).$$

- The hyperbolic tangent, cotangent, secant and cosecant are defined as follows:

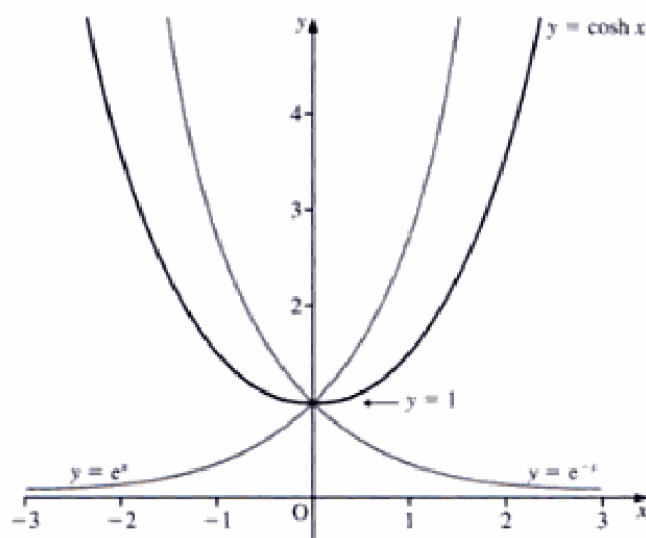
$$\tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

$$\coth x = \frac{1}{\tanh x} = \frac{e^x + e^{-x}}{e^x - e^{-x}}$$

$$\operatorname{sech} x = \frac{1}{\cosh x} = \frac{2}{e^x + e^{-x}}$$

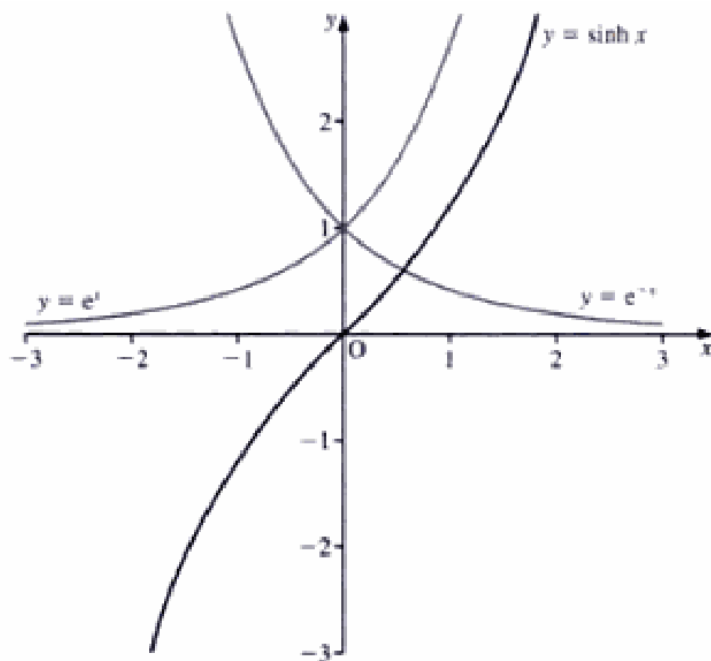
$$\operatorname{cosech} x = \frac{1}{\sinh x} = \frac{2}{e^x - e^{-x}}$$

- We obtain the graph of $y = \cosh x$ by finding the mean values of a few corresponding pairs of values of $y = e^x$ and $y = e^{-x}$, and then plotting these mean values.





- We obtain the graph of $y = \sinh x$ by finding the half the differences between a few corresponding pairs of values of $y = e^x$ and $y = e^{-x}$, and then plotting these values.



- Note that for both $\sinh x$ and $\cosh x$:
 - Both functions are continuous.
 - $\cosh x \geq 1$ for all values of x .
 - $\sinh x \rightarrow \cosh x$ as $x \rightarrow \infty$.
 - $\sinh x \rightarrow -\cosh x$ as $x \rightarrow -\infty$.
 - The graph of $\cosh x$ has a minimum point.
 - The graph of $\sinh x$ has no turning point.

- Since,

$$\tanh x = \frac{e^x - e^{-x}}{e^x + e^{-x}} = \frac{1 - e^{-2x}}{1 + e^{-2x}}$$

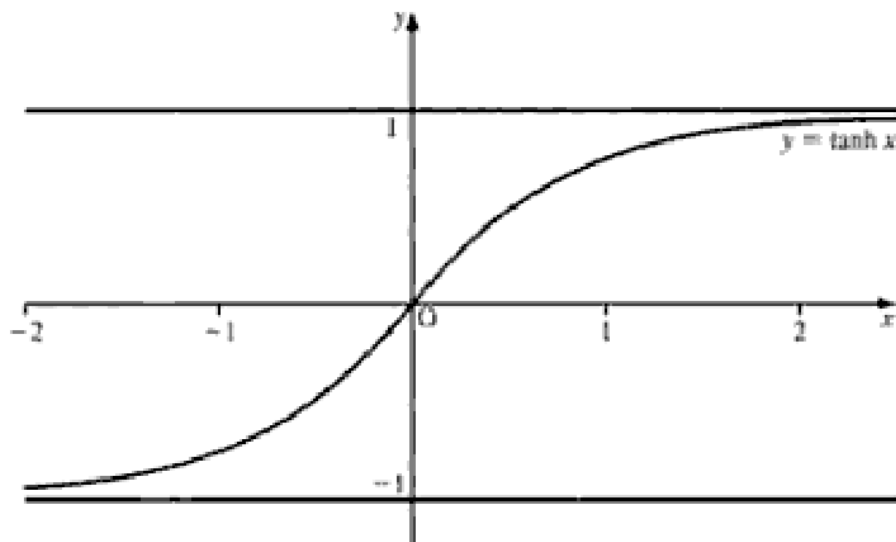
- Therefore, $\tanh x < 1$ for all values of x , and as $x \rightarrow +\infty$, $\tanh x \rightarrow 1$.

- Also,

$$\tanh x = \frac{e^x - e^{-x}}{e^x + e^{-x}} = -\frac{1 - e^{2x}}{1 + e^{2x}}$$

- Therefore, $\tanh x > -1$ for all values of x , and as $x \rightarrow -\infty$, $\tanh x \rightarrow -1$.

- Hence, the graph of $y = \tanh x$ lies between the asymptotes $y = 1$ and $y = -1$.



- Note that the similarity between hyperbolic and circular functions is limited to their occurrence in identities. The graphs have no corresponding relationship because, whilst all circular functions are periodic, none of the hyperbolic function is periodic.

Hyperbolic Identities

$$\cosh^2 x - \sinh^2 x \equiv 1$$

$$1 - \tanh^2 x \equiv \operatorname{sech}^2 x$$

$$\coth^2 x - 1 \equiv \operatorname{cosech}^2 x$$

$$\sinh(x \pm y) \equiv \sinh x \cosh y \pm \cosh x \sinh y$$

$$\cosh(x \pm y) \equiv \cosh x \cosh y \pm \sinh x \sinh y$$

$$\tanh(x \pm y) \equiv \frac{\tanh x \pm \tanh y}{1 \pm \tanh x \tanh y}$$

$$\sinh 2x \equiv 2 \sinh x \cosh x$$

$$\cosh 2x \equiv \cosh^2 x + \sinh^2 x \equiv 2 \cosh^2 x - 1 \equiv 1 + 2 \sinh^2 x$$

$$\tanh 2x \equiv \frac{2 \tanh x}{1 + \tanh^2 x}$$

- **Example 1: Prove the following identities from the basic definitions of the hyperbolic function.**

- (i) $\cosh^2 x - \sinh^2 x \equiv 1$
- (ii) $\cosh^2 x + \sinh^2 x \equiv \cosh 2x$
- (iii) $\sinh 2x \equiv 2 \sinh x \cosh x$
- (iv) $\sinh(x + y) \equiv \sinh x \cosh y + \cosh x \sinh y$
- (v) $\cosh(x + y) \equiv \cosh x \cosh y + \sinh x \sinh y$



Sunway A Levels

Osborn's Rule

- There is a striking similarity between the identities connecting hyperbolic functions and those connecting the corresponding circular functions.
- In fact the standard identities are in the same form except that certain signs are changed. Osborn's rule provides a simple way of remembering these changes of signs.
- Osborn's rule states that: To convert a standard trigonometrical identity into a hyperbolic one, replace $\cos \theta$ by $\cosh \theta$, and $\sin \theta$ by $\sinh \theta$. Then when two $\sinh x$ are multiplied together, multiply the product by -1 .
- This rule should be used with care as there are a few ways in which a product of two sine ratios can be disguised. E.g. in $\tan^2 x, \cot^2 x, \operatorname{cosec}^2 x$.

Warning:

- Osborn's rule holds for the standard trigonometrical identities but it must not be applied indiscriminately. For instance, application to

$$\frac{\cos 2x}{\cos x + \sin x} \equiv \cos x - \sin x \quad \text{and} \quad \frac{\cos x}{\sin y} - \frac{\sin x}{\cos y} \equiv \frac{2 \cos(x + y)}{\sin 2y}$$

leads to incorrect results.

- Osborn's rule cannot be relied upon as an aid to remembering calculus formulae.
- While Osborn's rule can be used to find a hyperbolic identity, it cannot be used to prove that identity. Proof is usually based, as in the above examples, on the definitions of the hyperbolic functions.

Exercise A:

1. Prove the following identities from the basic definitions of the hyperbolic function.

(i) $1 - \tanh^2 x \equiv \operatorname{sech}^2 x$

(ii) $\coth^2 x - 1 \equiv \operatorname{cosech}^2 x$

(iii) $\tanh 2x \equiv \frac{2 \tanh x}{1 + \tanh^2 x}$

(iv) $\cosh 3x \equiv 4 \cosh^3 x - 3 \cosh x$

(v) $\tanh x \equiv \sqrt{\frac{\cosh 2x - 1}{\cosh 2x + 1}}, x > 0$

(vi) $\cosh 3x \cosh^3 x + \sinh 3x \sinh^3 x \equiv \cosh^3 2x$



Sunway A Levels



2. If $\sinh x = \frac{3}{4}$, calculate $\operatorname{sech} x$, $\tanh x$, $\cosh 2x$ and $\tanh 2x$.
3. If $\tanh x = \frac{5}{13}$, calculate $\operatorname{cosech} x$, $\cosh x$, $\sinh 2x$ and $\tanh 2x$.

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4. Sketch the graph of $\coth x$, $\operatorname{cosech} x$, $\operatorname{sech} x$, $\cosh x + \sinh x$, $\sinh^2 x$, $\tanh(-x)$, $\sinh(-x)$.

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Sunway A Levels

To Solve Hyperbolic Functions

- To solve equations involving hyperbolic functions, can make use of the identities proved in the previous section, as well as the basic definitions of the hyperbolic functions.
- **Example 2: Solve the following hyperbolic equations.**

(i) $4 \cosh x - \sinh x = 5$

(ii) $12 \cosh^2 x + 7 \sinh x = 24$



(iii) $\cosh 2x - 7 \cosh x + 7 = 0$

(iv) $\sinh 2x + \cosh x = 0$

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Exercise B:

1. Solve the following hyperbolic equations.

(i) $11 \sinh x - 13 \cosh x = -7$

(ii) $6 \tanh x - 2 \operatorname{sech} x = 4$

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(iii) $3\operatorname{cosech} x + 5 = 4 \coth x$

(iv) $3 \sinh^2 x - \cosh x = 1$

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(v) $4 \sinh x - \cosh^2 x = 3$

(vi) $2 \operatorname{sech}^2 x - 3 \tanh x = 1$

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(vii) $\cosh 2x - 16 \sinh x = 4$

(viii) $2 \operatorname{sech} x + 119 = 121 \tanh x$

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2. Express $4 \cosh x + 5 \sinh x$ in the form $r \sinh(x + y)$ giving the values of r and $\tanh y$.
3. By expressing $13 \cosh x + 5 \sinh x$ in the form $r \cosh(x + y)$, find its minimum value.

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4. Prove that $\tanh(x + y) = \frac{\tanh x + \tanh y}{1 + \tanh x \tanh y}$.

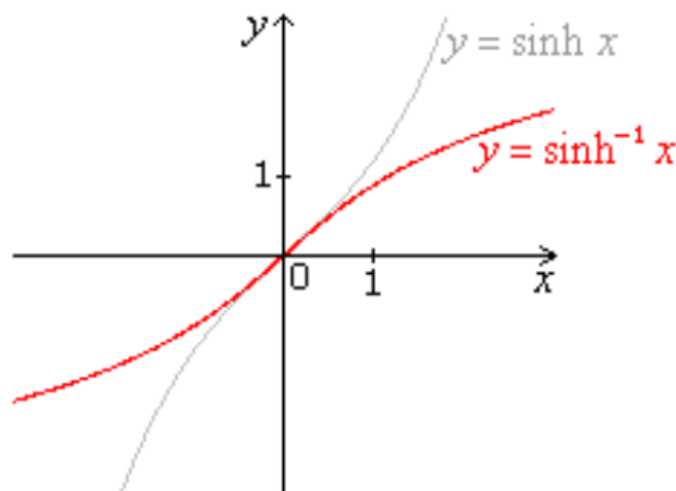
Deduce that $\tanh 3x = \frac{3 \tanh x + \tanh^3 x}{1 + 3 \tanh^2 x}$. Hence solve the equation $\tanh 3x = 2 \tanh x$.

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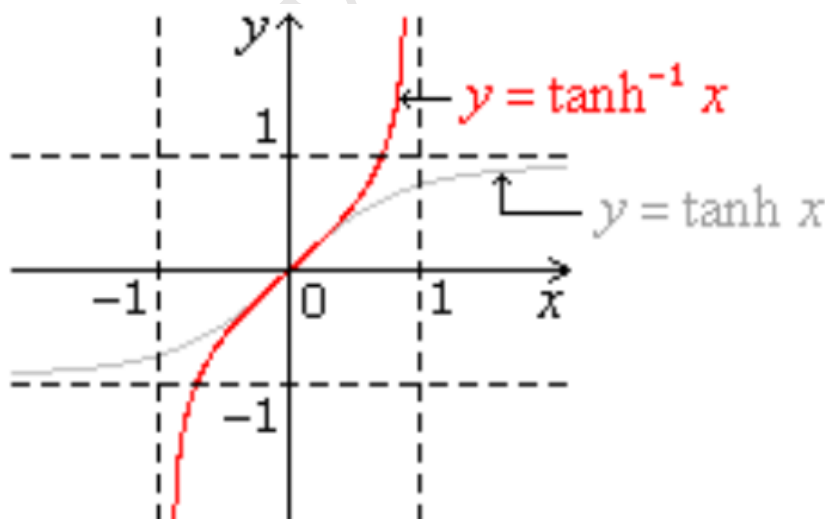


Inverse Hyperbolic Functions

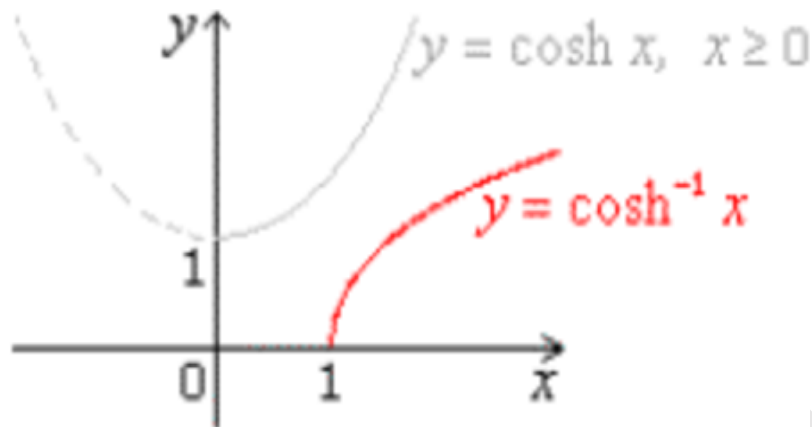
- The function $y = \sinh x$ is a one-to-one function with both domain and range equal to $\mathbb{R}$. Hence, it has an inverse function $y = \sinh^{-1} x$ which also has both domain and range equal to $\mathbb{R}$.



- The function $y = \tanh x$ is also one-to-one function and has domain $\mathbb{R}$ and range $-1 < y < 1$. Thus, it has an inverse function $y = \tanh^{-1} x$ with domain $-1 < x < 1$ and range $\mathbb{R}$.



- We have to more careful when defining inverse function to $y = \cosh x$, since it is not one-to-one function. By restricting the domain to $x \geq 0$, range would be $y \geq 1$. Then we have an inverse function $y = \cosh^{-1} x$ with domain $x \geq 1$ and range $y \geq 0$.



Inverse Hyperbolic Functions in terms of Logarithms

- Expressing $y = \cosh^{-1} x$ as a logarithmic function.

$$\text{Let } \cosh^{-1} x = y$$

$$x = \cosh y \Rightarrow x = \frac{1}{2}(e^y + e^{-y})$$

$$e^{2y} - 2xe^y + 1 = 0$$

$$e^y = \frac{2x \pm \sqrt{4x^2 - 4}}{2} = x \pm \sqrt{x^2 - 1}$$

$$y = \ln(x \pm \sqrt{x^2 - 1}) \quad \text{or} \quad y = \pm \ln(x + \sqrt{x^2 - 1})$$

$$\cosh^{-1} x = \ln(x + \sqrt{x^2 - 1}) \quad \text{for } x \geq 1, y \geq 0$$

- Expressing $y = \sinh^{-1} x$ as a logarithmic function.

$$\text{Let } \sinh^{-1} x = y$$

$$x = \sinh y \Rightarrow x = \frac{1}{2}(e^y - e^{-y})$$

$$e^{2y} - 2xe^y - 1 = 0$$

$$e^y = \frac{2x \pm \sqrt{4x^2 + 4}}{2} = x \pm \sqrt{x^2 + 1}$$

$$y = \ln(x \pm \sqrt{x^2 + 1})$$

$$y = \ln(x + \sqrt{x^2 + 1}), \text{ since } x < \sqrt{x^2 + 1}$$

$$\sinh^{-1} x = \ln(x + \sqrt{x^2 + 1})$$

- Expressing $y = \tanh^{-1} x$ as a logarithmic function.

Let $\tanh^{-1} x = y$

$$x = \tanh y \Rightarrow x = \frac{e^y - e^{-y}}{e^y + e^{-y}}$$

$$e^{2y}(1 - x) = 1 + x$$

$$e^{2y} = \frac{1 + x}{1 - x}$$

$$y = \frac{1}{2} \ln \left(\frac{1 + x}{1 - x} \right)$$

$$\tanh^{-1} x = \frac{1}{2} \ln \left(\frac{1 + x}{1 - x} \right), |x| < 1$$

- **Practice:**

Expressing $y = \operatorname{sech}^{-1} x$, $y = \operatorname{cosech}^{-1} x$, $y = \operatorname{coth}^{-1} x$ as a logarithmic function.



Sunway A Levels

To Solve Inverse Hyperbolic Equations

- To solve equations involving inverse hyperbolic functions, can make use of the identities proved in the previous section, as well as of the logarithmic forms of the functions.

- **Example 3:**

Solve the equation $\sinh^{-1} x = \ln(2 - x)$, giving your answer correct to 3 decimal places.

- **Example 4:**

Prove that $\sinh(A - B) \equiv \sinh A \cosh B - \cosh A \sinh B$.

Solve the equation $\sinh^{-1} 2x = \sinh^{-1} \left(\frac{3}{4} \right) - \sinh^{-1} x$.

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- **Example 5:**

Solve the equation $\tanh^{-1} 2x = \tanh^{-1} x + \tanh^{-1} \left(\frac{3}{7}\right)$.

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- **Example 6:**

Prove that $\sinh^{-1} x = \ln(x + \sqrt{x^2 + 1})$, and deduce that $\operatorname{cosech}^{-1} x = \ln\left(\frac{1 + \sqrt{x^2 + 1}}{x}\right)$.

Find the integer n such that $\ln n = \sinh^{-1}\left(\frac{3}{4}\right) + \operatorname{cosech}^{-1}\left(\frac{3}{4}\right)$.

Exercise C

1. Find solutions to the following equations expressing the solutions in logarithmic form.

(i) $2 \cosh^2 x - 5 \sinh x = -1$

(ii) $\sinh^2 x - 6 \cosh x + 9 = 0$

(iii) $6 \operatorname{sech}^2 x + \tanh x = 5$

(iv) $12 \tanh^2 x + 11 \operatorname{sech} x = 14$

(v) $\sinh^{-1} 3x = \cosh^{-1}(5x)$

(vi) $\cosh^{-1}(13x) = \sinh^{-1}(5x)$

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2. (a) Using the logarithm form, show that $\tanh^{-1} x + \tanh^{-1} y = \tanh^{-1} \left(\frac{x+y}{1+xy} \right)$.
- (b) Find a similar expression for $\tanh^{-1} x - \tanh^{-1} y$.

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3. Express each of the following in logarithm form.

(i) $\operatorname{sech}^{-1} x$

(ii) $\operatorname{cosech}^{-1} x$

(iii) $\tanh^{-1} x$

(iv) $\sinh^{-1} \left(\frac{1}{x} \right)$

(v) $\cosh^{-1}(1 - x^2)$

(vi) $\tanh^{-1} \left(\frac{1}{x} \right)$